YIYAO ZHU

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Academic Appointments

National University of Singapore, Department of Economics
Postdoctoral Research Fellow
Postdoctoral Mentor: Yingkai Li
2025–Present

Education

National University of Singapore
Ph.D. in Economics
Dissertation: *Essays on Contests*
Ph.D. Advisor: Jingfeng Lu
2020–2025

National University of Singapore
B.Sc. (Hons.) in Applied Mathematics
Specialization: Operations Research and Financial Mathematics
Second Major: Economics
2016–2020

Fields

Microeconomic Theory (Contest Theory and Mechanism Design)

References

Jingfeng Lu
Professor / Provost's Chair
Department of Economics
National University of Singapore
ecsljf@nus.edu.sg

Yingkai Li
Assistant Professor (Presidential Young Professor)
Department of Economics
National University of Singapore
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Satoru Takahashi
Lim Chong Yah Professor
Department of Economics
National University of Singapore
ecsst@nus.edu.sg

Research Papers

Optimal Public Programs: Transfers versus Ordeals

with Yingkai Li

Abstract. We study the optimal design of public programs when scarce goods are allocated using either monetary transfers or costly ordeals. With risk-averse agents, the two screening instruments generate sharply different mechanisms. Under costly ordeals, an optimal mechanism takes a simple two-ticket form: intermediate types receive a rationed standard option, while high types obtain guaranteed service at a higher cost. Under monetary transfers, allocation probabilities vary more finely with type, with payments characterized by a common price wedge and endogenous entrance fees.

Delegated Prize Design in Team Contests

with Hideo Konishi and Jingfeng Lu

Abstract. We study prize design in team contests when a designer controls team-level rewards but delegates their internal allocation to coaches. For any operational rule, the cost-efficient effort continuation reduces to an asymmetric Tullock contest in which the internal allocation enters through a scalar virtual unit cost. With winning-only prizes, the designer and each coach select virtual-cost-minimizing rules. Optimal direct design can be delegated under public rules and, with passive beliefs, under private rules. Outcome-contingent prizes separate incentives from rents: win-lose spreads implement effort, while transfer levels extract surplus. First-best spreads are teamwise uniform, and full extraction requires negative losing transfers. When rules are public and any member can veto the contest, rival participation constraints discipline each coach, so delegated team budgets implement the first best. A two-team CES example shows that observability is substantive: under private rules and the passive-belief restriction, no pure-strategy continuation equilibrium implements the first best.

Dividing the Spoils: Strategic Prize Allocation in Team Contests *Job Market Paper*
with Zhonghong Kuang and Jingfeng Lu

Abstract. Rival teams compete through battles and reward members only after team victory. We study how team managers allocate victory-contingent reward budgets across players to maximize their teams' winning probabilities in a majoritarian contest. Under a regularity condition on battle technologies, the allocation game has a unique pure-strategy equilibrium. Despite differences in budgets and player costs across teams and technologies across battles, both managers choose the same normalized reward schedule, a property we call reward schedule alignment. Each battle's share is proportional to discriminatory power, closeness, and pivotality. With precommitted rewards, allocations and team-winning probabilities are independent of battle order.

Move Orders in Contests: Optimal Handicaps and Effort Ranking
with Lei Gao, Jingfeng Lu, and Zhewei Wang

Abstract. We study how a contest designer should jointly choose identity-contingent favoritism and move order in a two-player generalized Tullock contest to maximize expected total effort. We show that, when favoritism can be chosen freely, weak-player leadership is never optimal under cost asymmetry. When contest accuracy is relatively low, strong-player leadership is optimal. If the players are sufficiently similar in ability, the designer favors the strong leader to reinforce its commitment incentive; if the ability gap is sufficiently large, the designer instead favors the weak follower to level the playing field and preserve competitive pressure. When contest accuracy is sufficiently high, the optimal strong-lead contest is preemptive and, under cost asymmetry, favors the weak follower, thereby forcing the strong leader to exert more effort to deter entry. Simultaneous play can be optimal only when contest accuracy is high and the players are extremely similar in ability.

Where Teams Randomize: Multi-Task Team All-Pay Contests
with Jingfeng Lu

Abstract. We study a complete-information all-pay contest between two teams performing $K \geq 2$ complementary tasks under Cobb–Douglas aggregation, and we completely classify *one-pair mixed* equilibria, in which exactly one member of each team randomizes. Only maximal-weight tasks can host randomization, and every ordered pair of such tasks, with diagonal pairs denoting same-task mixing, supports exactly one equilibrium. Hence there are $|\mathcal{K}^*|^2$ labeled equilibria, where $\mathcal{K}^*$ is the set of maximal-weight tasks. All of them induce identical member-by-member payoffs, expected effort costs, and winning probabilities, governed by a single effective cost ratio. No member earns less than in the all-zero equilibrium; unless every task has maximal weight and the effective cost ratio equals one, at least one member earns strictly more. Each equilibrium path can be implemented under block-sequential timing, with unchanged on-path play, exactly when all randomizers move simultaneously in the terminal block; this timing result uses a positive-history continuation concept.

Seminars and Conferences

2025 NUS Micro Theory Lunches (Singapore)

2024 NUS Micro Theory Lunches (Singapore); Asian Meeting of the Econometric Society (Hangzhou); Conference on Mechanism and Institution Design (Budapest); Asian Meeting of the Econometric Society (Ho Chi Minh City); East Asia Game Theory Conference (Jeju); China Meeting on Game Theory and Its Applications (Shenzhen); GAMES 2024 (Beijing)

2023 Asian Meeting of the Econometric Society (Singapore)

Awards and Fellowships

Research Scholarship, Department of Economics, National University of Singapore
2024–2025

Research Scholarship, Ministry of Education, Singapore
2023–2024

Graduate Teaching Fellowship, National University of Singapore
2020–2023

Science and Technology Undergraduate Scholarship, National University of Singapore
2016–2020

Teaching Experience

EC3102 Macroeconomic Analysis II, Teaching Assistant, National University of Singapore
2022–2024

Professional Service

Referee: *International Journal of Game Theory*; *The Japanese Economic Review*